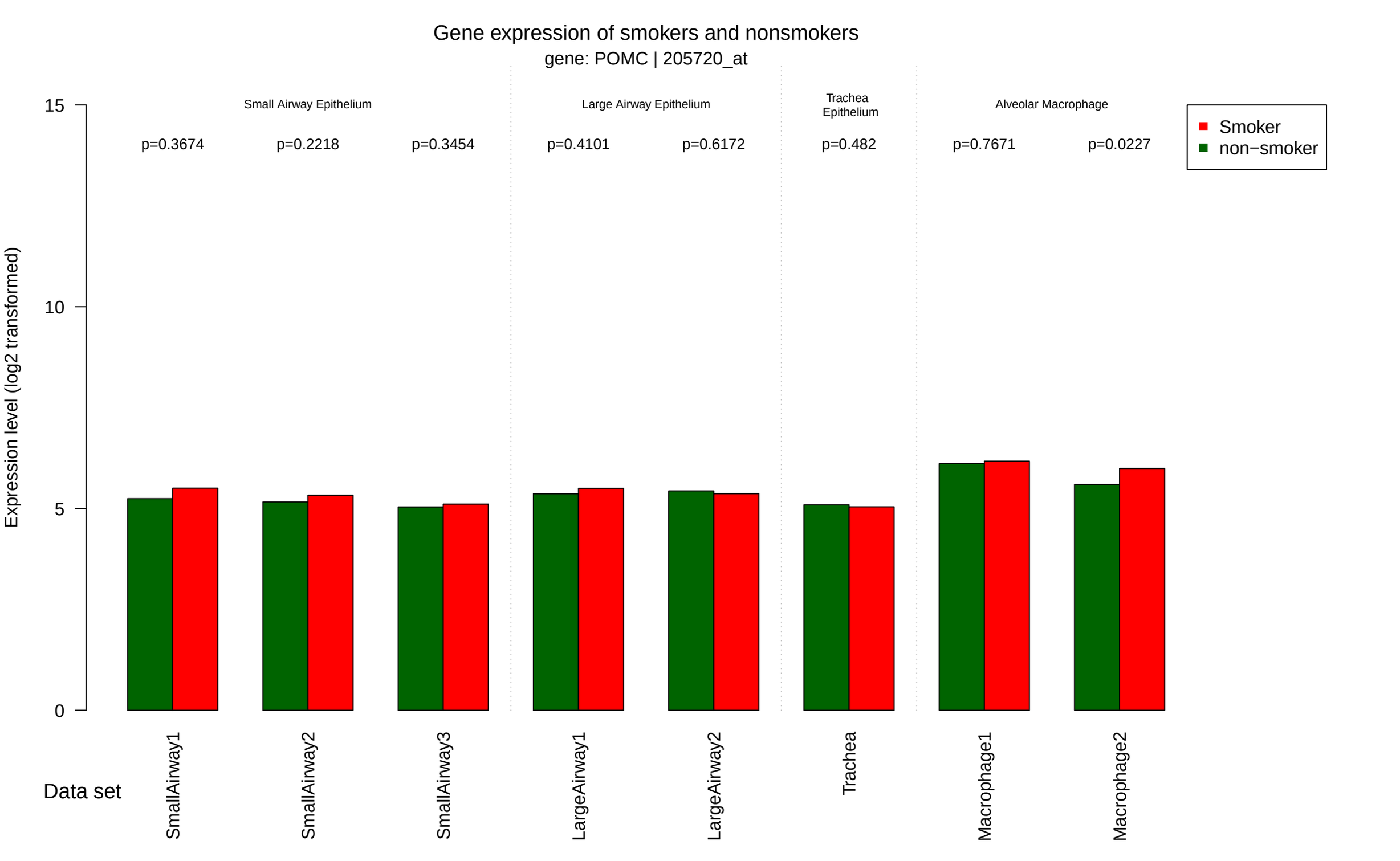
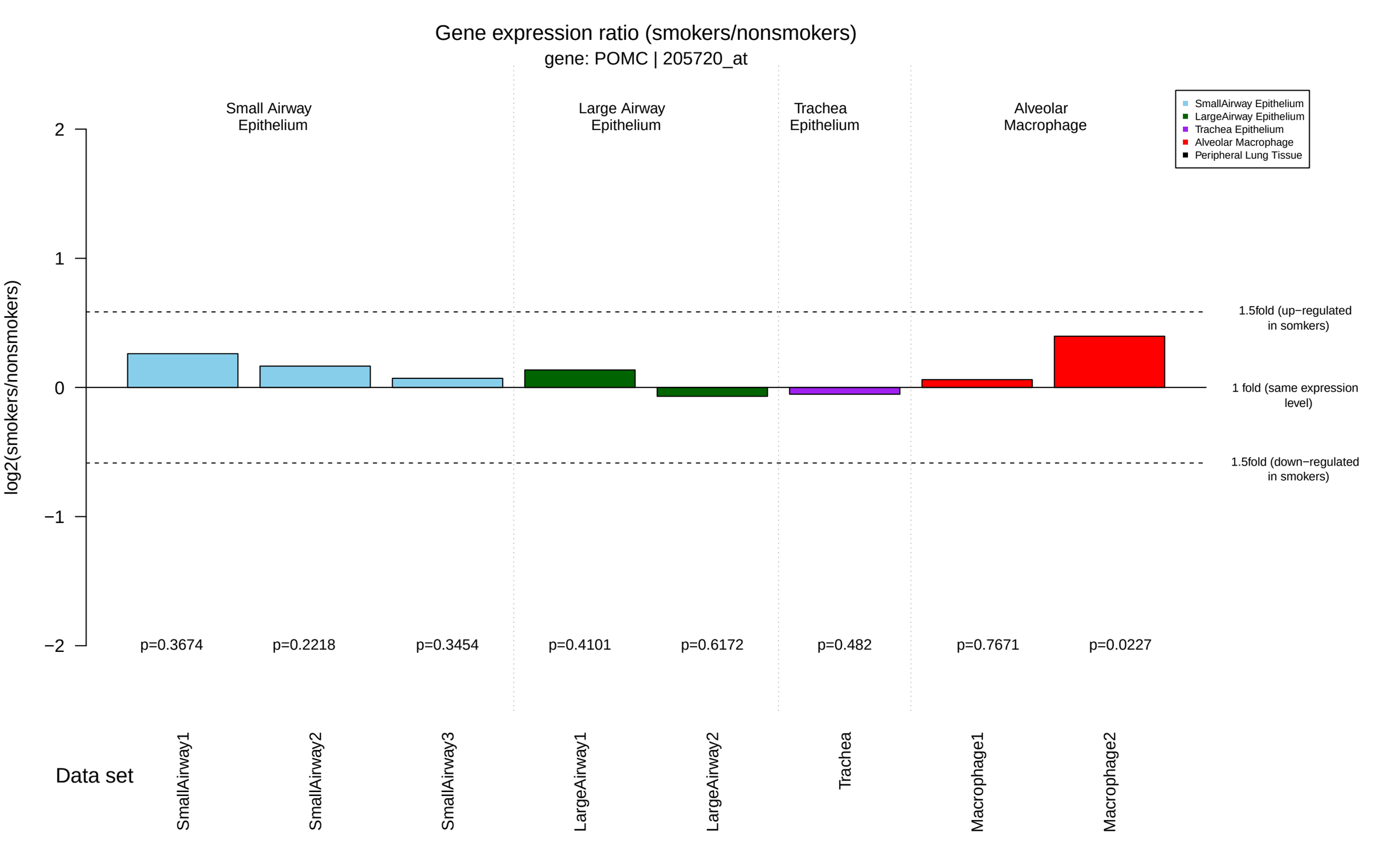
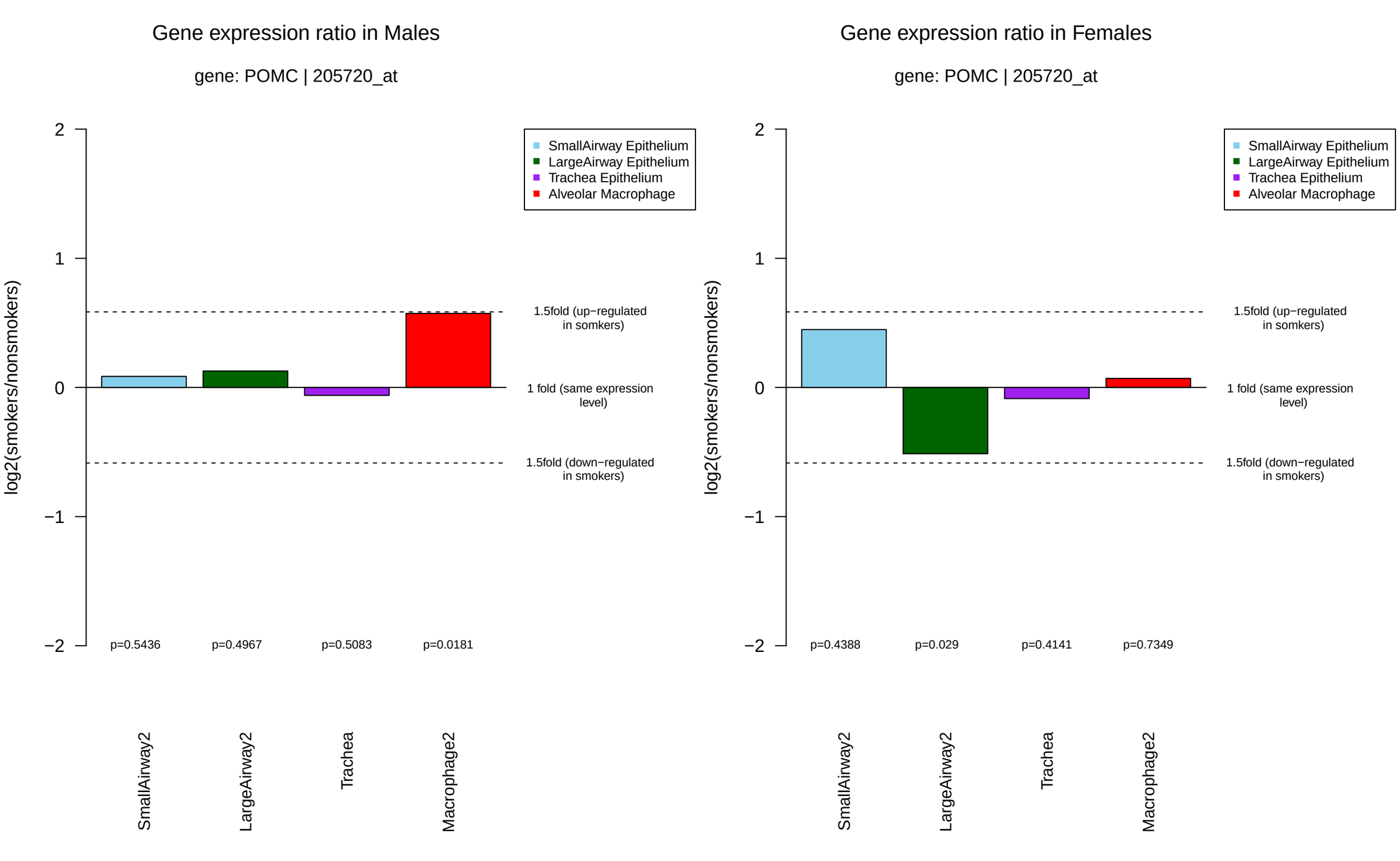
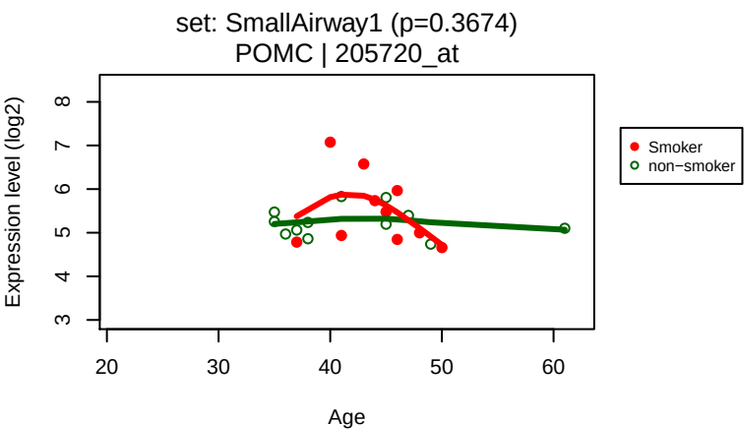
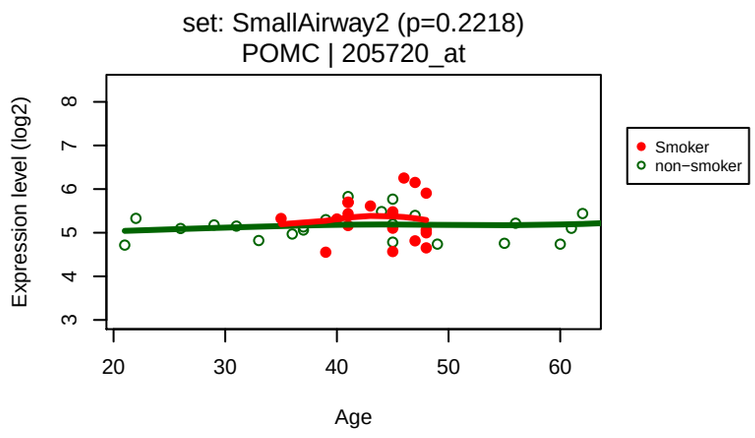
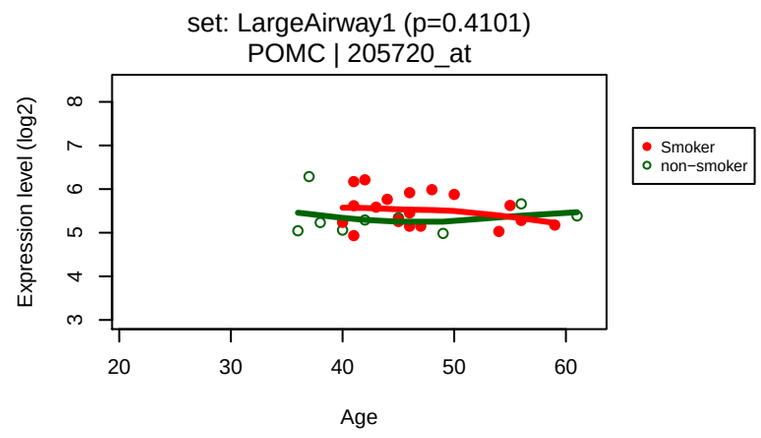

Gene expression of smokers and nonsmokers
gene: POMC | 205720_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3674
p=0.2218
p=0.3454
p=0.4101
p=0.6172
p=0.482
p=0.7671
p=0.0227
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

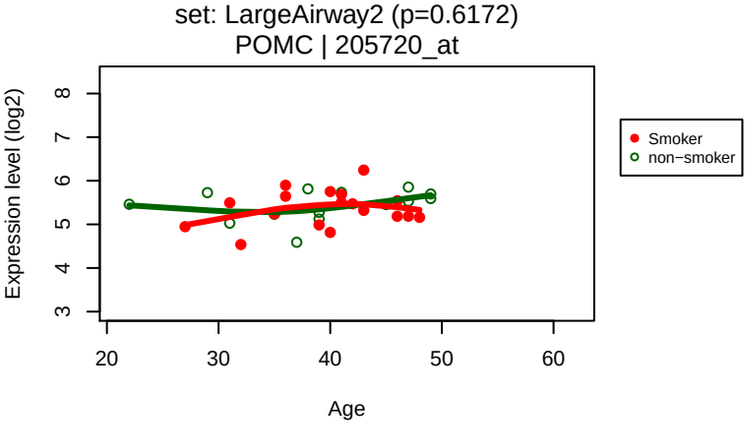

Gene expression ratio (smokers/nonsmokers)
gene: POMC | 205720_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3674
p=0.2218
p=0.3454
p=0.4101
p=0.6172
p=0.482
p=0.7671
p=0.0227
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

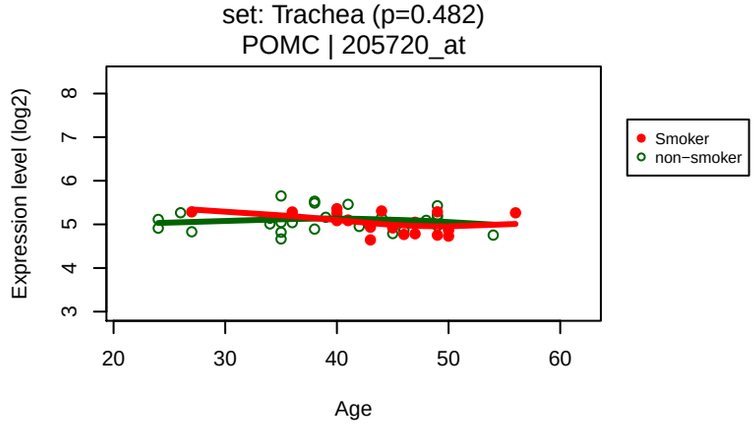

Gene expression ratio in Males
gene: POMC | 205720_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5436
p=0.4967
p=0.5083
p=0.0181
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POMC | 205720_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4388
p=0.029
p=0.4141
p=0.7349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

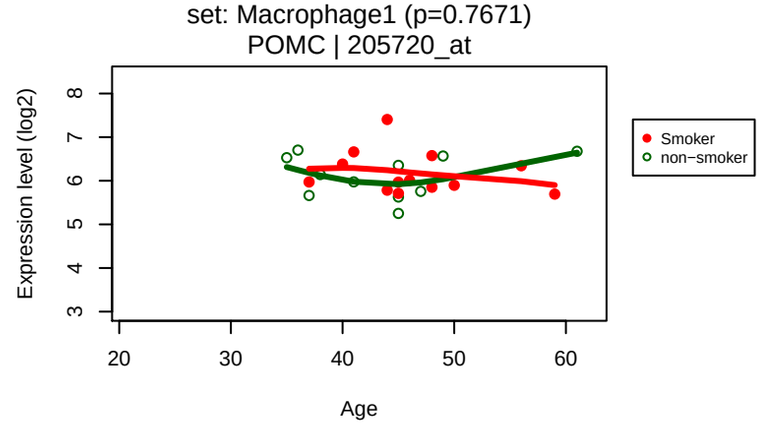

set: SmallAirway1 (p=0.3674)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker

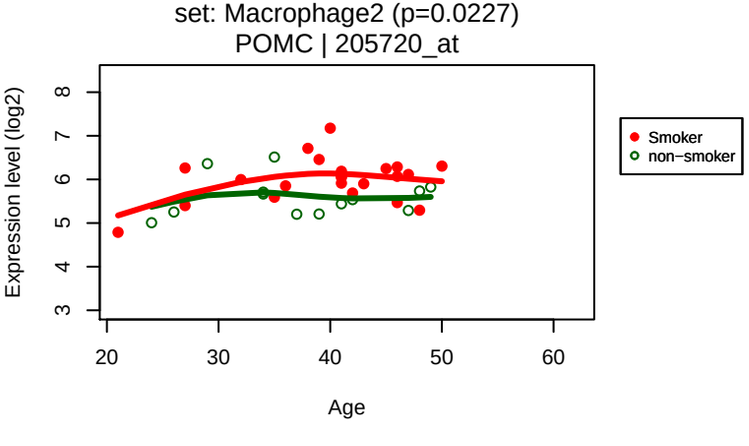

set: SmallAirway2 (p=0.2218)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker

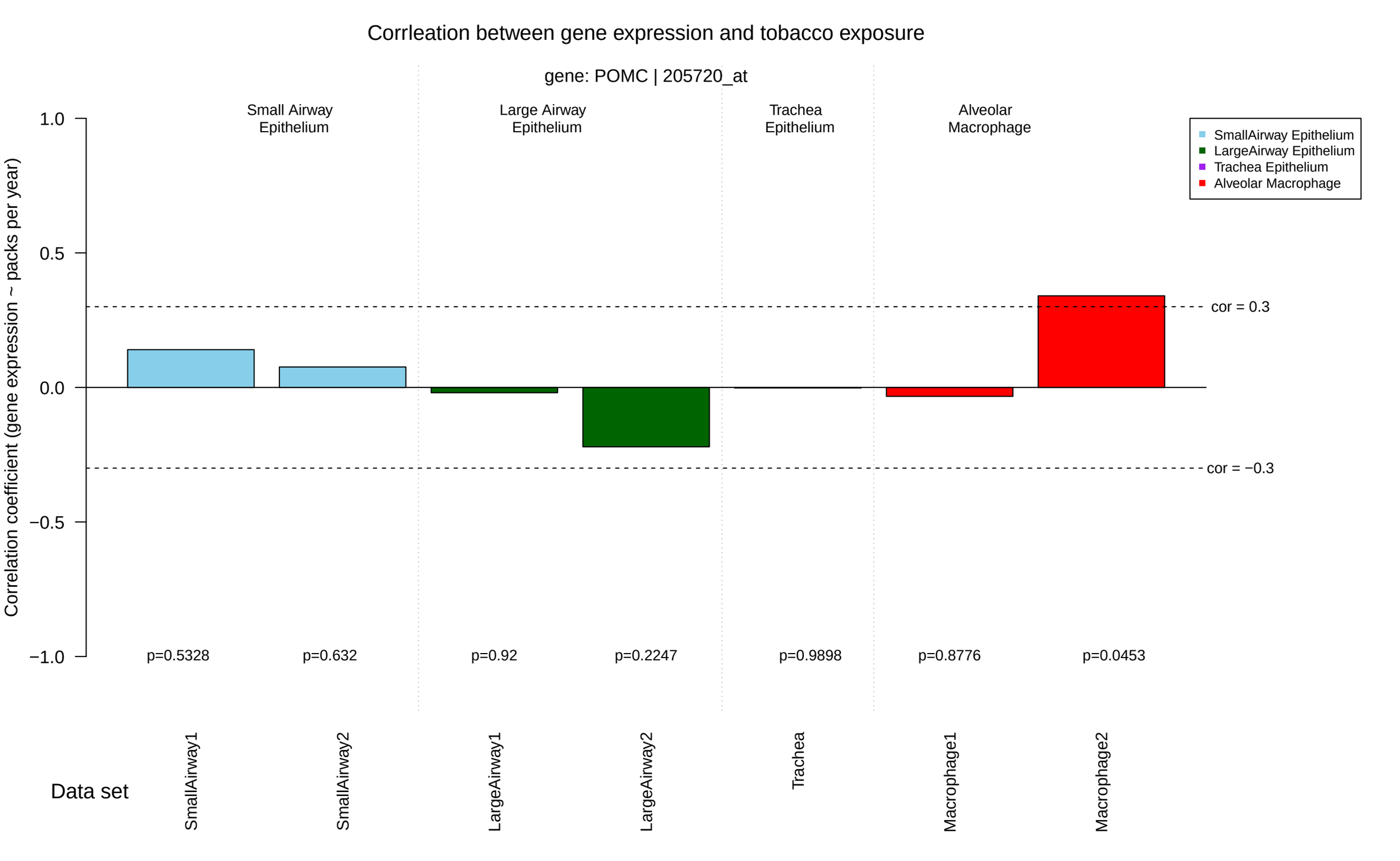

set: LargeAirway1 (p=0.4101)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6172)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.482)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7671)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0227)
POMC | 205720_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POMC | 205720_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5328
p=0.632
p=0.92
p=0.2247
p=0.9898
p=0.8776
p=0.0453
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2